

Network Diagram

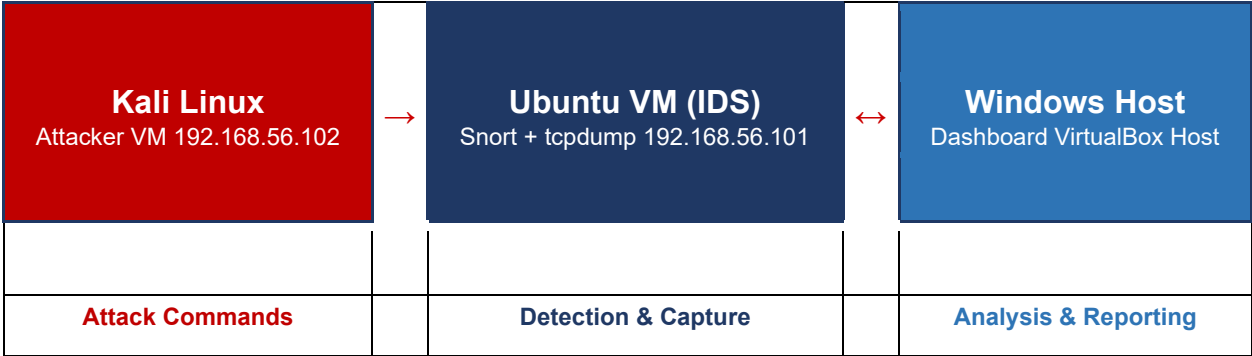
1. Network Overview

This document describes the virtual network infrastructure used for the Hybrid Intrusion Detection System (IDS) project. The lab environment simulates a realistic attacker-target scenario entirely within VirtualBox, with three participants: a Kali Linux attacker VM, an Ubuntu IDS target VM, and the Windows host that runs the ML analysis pipeline and dashboard.

All machines communicate over a VirtualBox Host-Only network (192.168.56.0/24), isolated from the real internet to ensure safe, controlled experimentation.

2. Network Topology Diagram

The diagram below illustrates the relationships and data flows between each component in the lab environment.



3. Component Details

3.1. Kali Linux – Attacker VM

Kali Linux is used exclusively as the attacker machine. It has no IDS tools installed. All attack commands are issued from this machine towards the Ubuntu VM's monitored interface.

Host-Only IP: 192.168.56.102

Role in lab: simulates an external attacker performing reconnaissance and network scanning.

3.2. Ubuntu VM – IDS target

Ubuntu is the monitored target machine. It runs both Snort (signature-based IDS) and tcpdump (packet capture) simultaneously on the enp0s8 interface. Logs are written to the VirtualBox shared folder /media/sf_snortlogs, which is accessible from the Windows host as snortlogs/. **Host-Only IP: 192.168.56.101**

Monitored Interface: enp0s8 (Host-Only)

3.3. Windows Host – Analysis & Dashboard

The Windows host runs the Python-based ML pipeline. It reads the Snort alert log and pcap file from the shared folder, runs the Random Forest model, and generates results CSVs. The Streamlit dashboard is served at <http://localhost:8501>.

Key files produced:

- hybrid_results.csv — ML verdict on each Snort alert
- pcap_analysis_results.csv — All pcap flows with ML classification of missed flows

4. Network Interfaces

Interface	Type	IP Address	Purpose
enp0s3	NAT (VirtualBox)	10.0.2.x (DHCP)	Internet access for Ubuntu VM
enp0s8	Host-Only Network	192.168.56.101	Attack surface — Snort & tcpdump monitor this
eth1 (Kali)	Host-Only Network	192.168.56.102	Attacker interface — all attack commands sent here
VirtualBox HostOnly	Virtual Switch	192.168.56.1 (Host)	Connects Kali, Ubuntu, and Windows Host

5. Data Flow

Step	Component	Action	Output
1	tcpdump (Ubuntu)	Captures ALL packets on enp0s8	traffic.pcap
2	Snort (Ubuntu)	Matches packets vs rules	alert log file
3	hybrid_pipeline.py (Windows)	Reads alert log, extracts features, runs ML	hybrid_results.csv
4	pcap_analyzer.py (Windows)	Reads pcap, compares vs Snort alerts, ML on missed flows	pcap_analysis_results.csv
5	dashboard.py (Windows)	Visualises results in Streamlit	http://localhost:8501

6. Attack Commands Used

Command	Attack Type	Detected By
ping -c 5 192.168.56.101	ICMP Echo — connectivity check	Snort (ICMP rule) + ML
nmap -sS 192.168.56.101	SYN (Stealth) Scan — half-open TCP scan	Snort + ML (high serror_rate)
nmap -sX 192.168.56.101	XMAS Scan — FIN, PSH, URG flags set	Snort (XMAS rule) + ML
nmap -sN 192.168.56.101	NULL Scan — no flags, evades some firewalls	Snort + ML (probe pattern)
nmap -A 192.168.56.101	Aggressive Scan — OS detect, service detect, scripts	Snort (partial) + ML (high confidence)

7. Monitoring Commands

7.1. Starting Snort

```
sudo snort -A fast -q -c /etc/snort/snort.conf -i enp0s8 -l /media/sf_snortlogs
```

- -A fast: writes alerts in fast (single-line) format
- -q: quiet mode — suppresses banner
- -c: specifies the Snort configuration file
- -i enp0s8: monitor the Host-Only interface
- -l /media/sf_snortlogs: write logs to the shared folder

7.2. Starting tcpdump

```
sudo tcpdump -i enp0s8 -w /media/sf_snortlogs/traffic.pcap
```

- Captures every packet on enp0s8 to a pcap file
- Must be started BEFORE the attack for full capture
- Stopped with Ctrl+C after the attack session ends

7.3. Viewing Results

```
cat /media/sf_snortlogs/alert
```

```
sudo tcpdump -r /media/sf_snortlogs/traffic.pcap
```

8. Network State: Before and After Testing

Before Testing

- No traffic between Kali and Ubuntu VMs
- Snort running but no alerts generated
- tcpdump running and capturing (near-empty pcap)
- Dashboard shows empty results

After Testing

- Snort alert log populated with matched rule events
- traffic.pcap contains full record of all packets sent and received
- hybrid_results.csv shows ML severity rating for each Snort alert
- pcap_analysis_results.csv shows flows Snort missed — ML classifies these as ATTACK, PROBE, or NORMAL
- Dashboard Tab 1 shows attack breakdown; Tab 2 shows Snort coverage gaps